| | |
|---|---|
| | Net work done on gas = 30 – 45 = – 15 J. |
| **13** | **D**<br>$v = \omega\sqrt{{x_0}^2 - \frac{{x_0}^2}{4}}$<br>$v = \frac{\sqrt{3}}{2}\omega x_0 = \frac{\sqrt{3}}{2}v_0$ |
| **14** | **D**<br>Air molecule at point P is moving in the positive x-direction. Refer to graph drawn |